

Tagline: *"Connecting People, Devices, and Possibilities"*

INTRODUCTION TO NETWORKING

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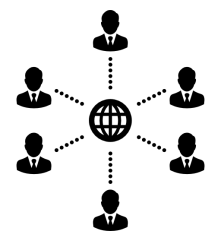
Cybersecurity C26 - 04: Class 3

Networking Fundamentals: Connecting the Digital World

Welcome to our Introduction to Networking class. In this session, we'll explore the **core concepts and structure of computer networks**, how devices communicate, key network types, essential components like routers and switches, and foundational models such as the **OSI and TCP/IP**. These principles form the backbone of today's **internet and digital communication**

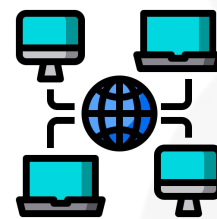
Learning Objectives

By the end of this session, participants will be able to:



Networking concepts

Explain key networking concepts (LAN, WAN, IP, etc.)



Network devices

Identify key network devices (switch, router, firewall).



OSI/TCP Model

Describe how data flows between devices using the OSI Model.



Cisco Packet Tracer

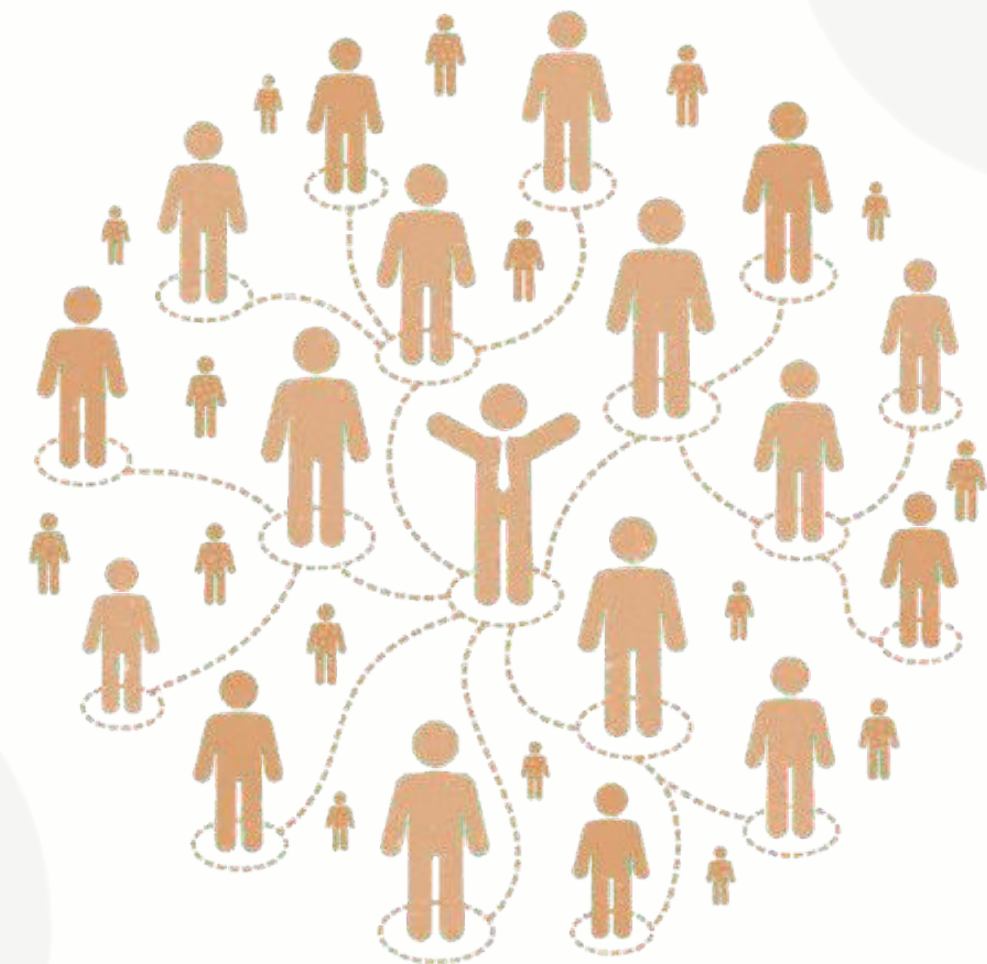
Configure a basic network in **Cisco Packet Tracer**.

What is Networking?

Networking is the practice of **connecting two or more computing devices** together to share data, resources, and services.

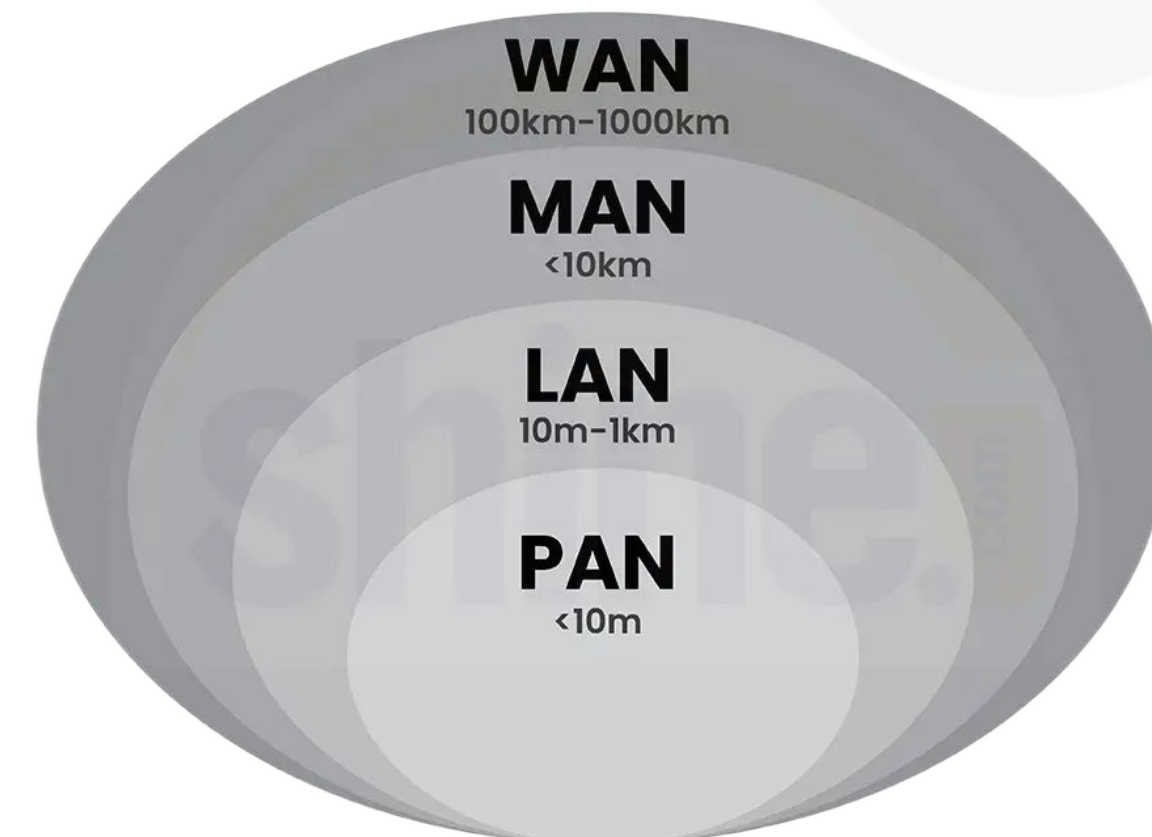
Key Purposes of Networking

- **Communication:** Send emails, chat, video call, or transfer files.
- **Resource Sharing:** Use shared printers, storage, and applications.
- **Data Access:** Retrieve data from servers or the cloud.
- **Centralized Management:** Control and secure multiple systems from one place.



Types of Networks

Type	Full Form	Description
PAN	Personal Area Network	Very small area, like around a person e.g Bluetooth
LAN	Local Area Network	Small area, like an office or school
WAN	Wide Area Network	Large area, e.g. the Internet
MAN	Metropolitan Area Network	Covers a city or campus



Network Topologies

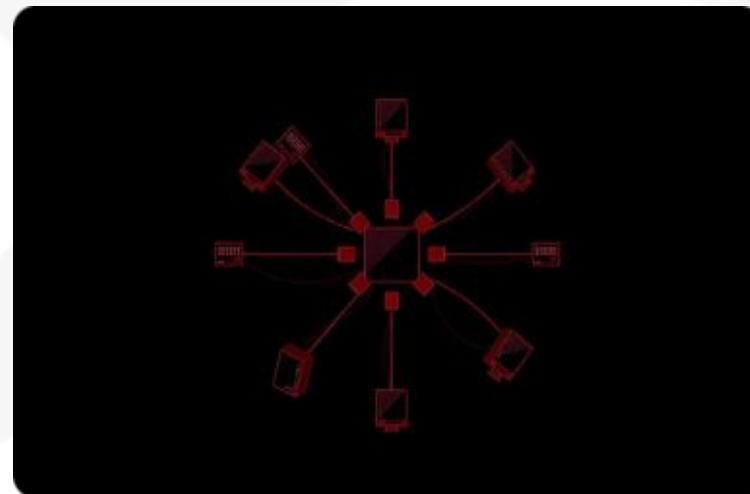


Bus Topology



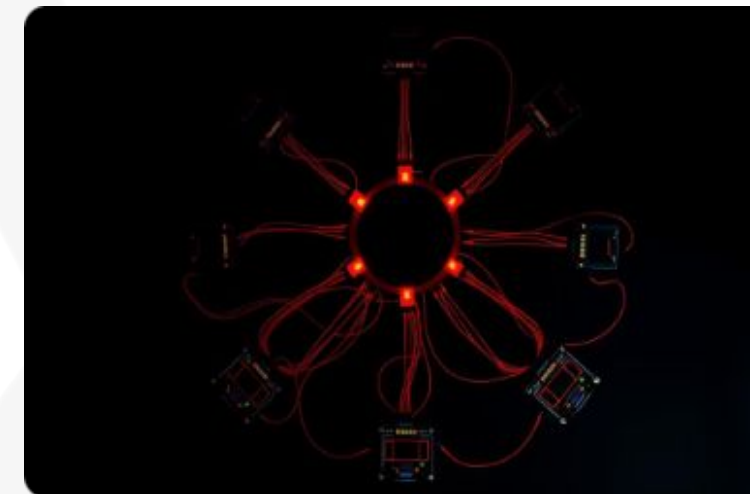
A straight line with multiple devices connected.

Star Topology



Devices connected to a central switch.

Ring Topology



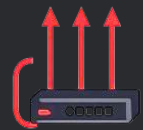
Devices connected in a loop.

Mesh Topology



Every device connected to every other device.

Networking Devices



Router: Connects different networks together (e.g., LAN to the internet).



Switch: Connects multiple devices within the same local network (LAN).



Gateway: Acts as a translator connecting two different types of networks (e.g., internal network to external).



Hub: Broadcasts data to all connected devices, used in basic or legacy networks.



Server: Provides services or resources (like files, DHCP, DNS) to client devices.



Firewall: Protects a network by filtering traffic and blocking unauthorized access while allowing safe communication between devices.



Understanding Common Network Protocols & Ports

Network protocols are **rules for how data is sent and received** across a network. Each uses a **specific port number** so devices know which service the data is for.



HTTP (80) - Used for standard web traffic (browsing websites).



HTTPS (443) - Secure web traffic with encryption (SSL/TLS).



FTP (20, 21) - Transfers files between client and server.



SSH (22) - Secure remote login and management of devices.



SMTP (25) - Sends emails from a client to a mail server.



DNS (53) - Translates domain names to IP addresses.



DHCP (67, 68) - Automatically assigns IP addresses to devices.

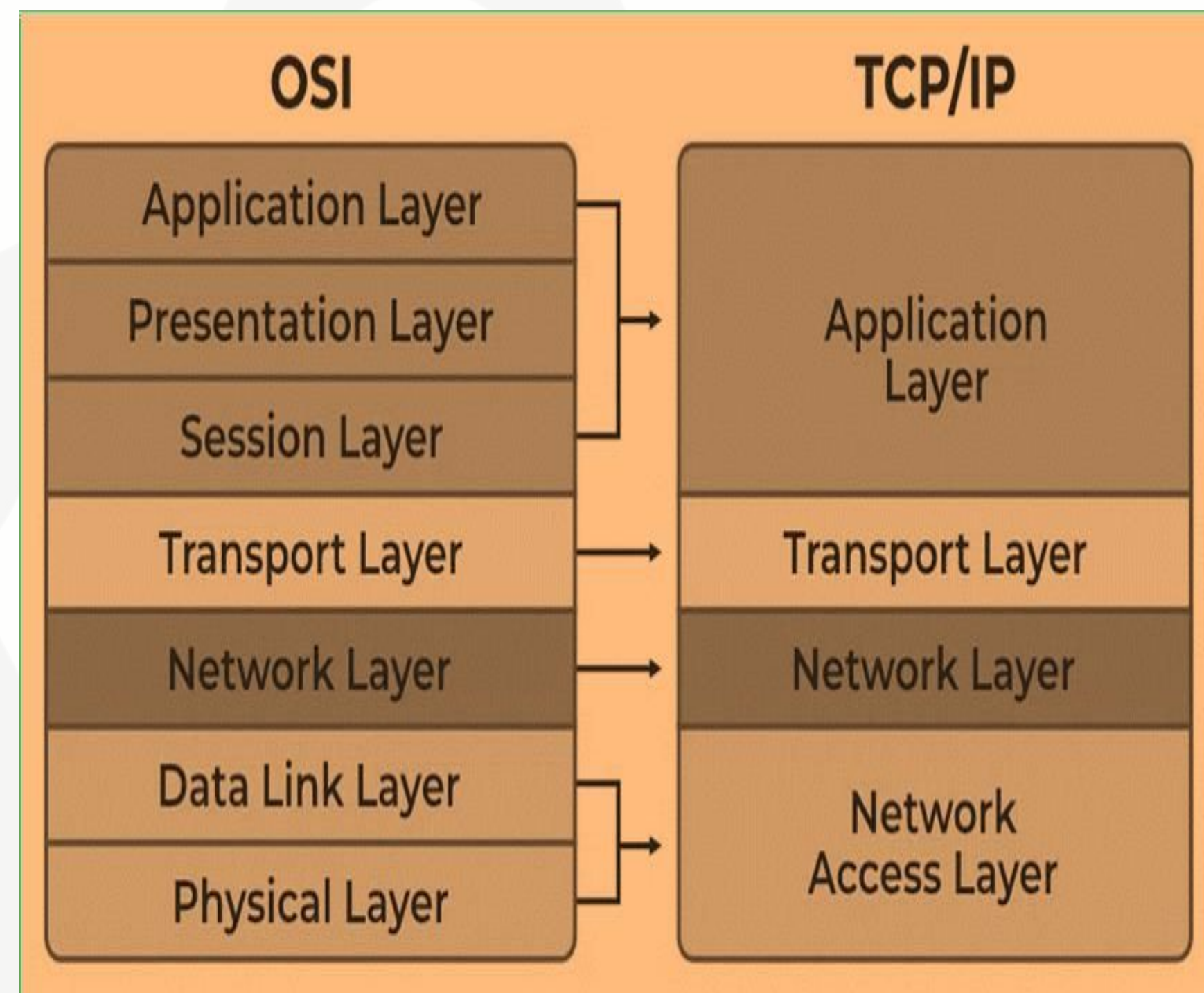


RDP (3389) - Connects to remote desktops (Windows Remote Desktop).

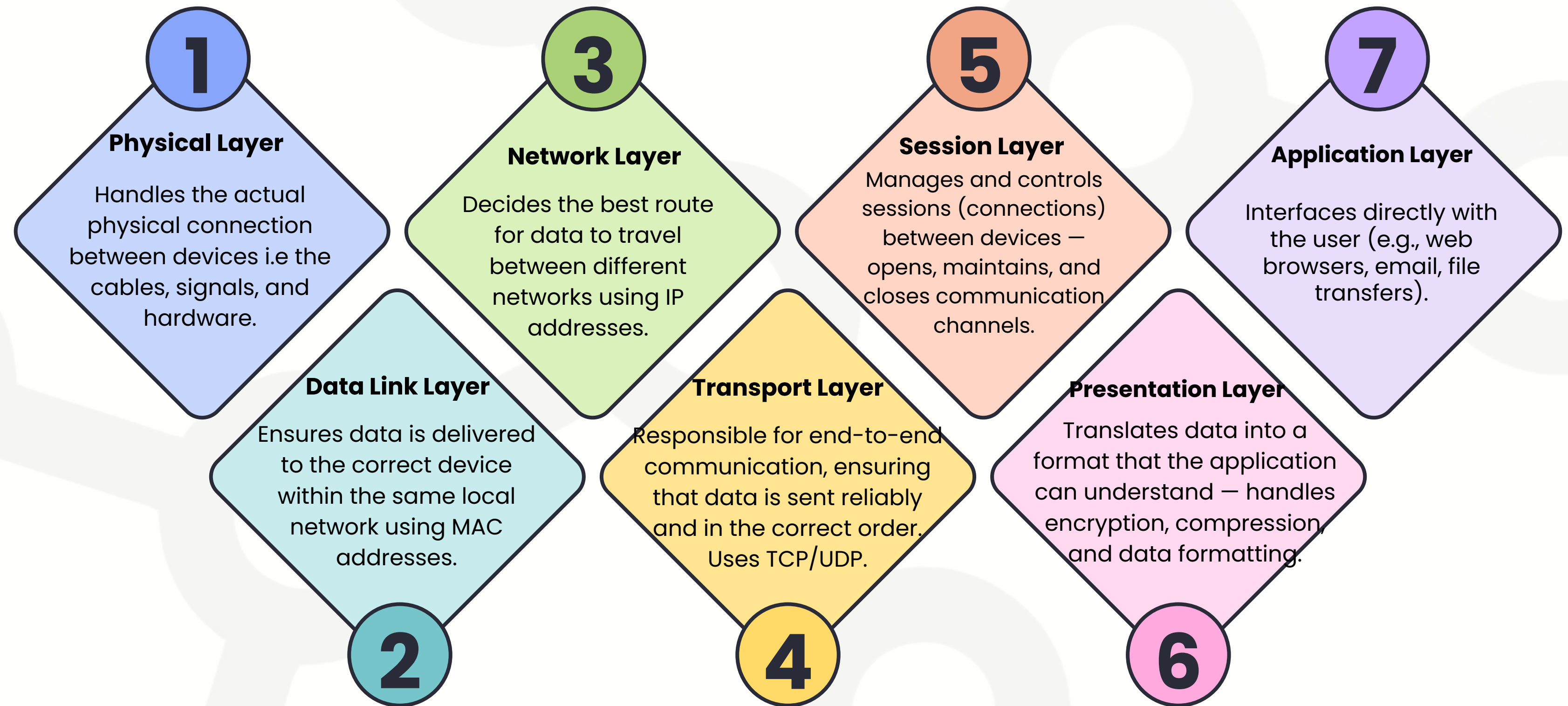
OSI and TCP/IP Models

OSI Model: A 7-layer framework ensuring seamless communication across networks. Each layer, from physical to Application, serves a unique function

TCP/IP Model: A simplified 4-layer protocol stack designed for real-world internet applications.



OSI and TCP/IP Models



Key Aspects of IP Addressing

Unique Identification: Every device on a network has a unique IP address to enable communication.

Types of IP

- **IPv4** (e.g. 192.168.1.1) — most common format.
- **IPv6** (e.g. 2001:0db8::1) — newer version with more addresses.

Private vs Public

- **Private IPs** are used within local networks.
- **Public IPs** are used on the internet.

Static vs Dynamic

- **Static IP:** Manually assigned, stays the same.
- **Dynamic IP:** Automatically assigned by a DHCP server.

Subnet Mask: Defines the network and host portions of an IP address (e.g. 255.255.255.0).

Specialization:

Network Design & Security

Business Focus:Secure Network Infrastructure for Small
Businesses**Skill Outcome:**

- Network design and basic LAN setup
- IP addressing and device configuration
- Connectivity testing and troubleshooting

DATA BRIDGE INNOVATIONS**Building a SOHO Network: Introduction to
Networking and Network Security****Project Learning Opportunities:**

- Learn how to design and configure a small routed network.
- Understand basic IP addressing and inter-network communication.
- Gain hands-on experience using Cisco Packet Tracer for network simulation.

Tools and Technology Used:

Company Introduction

You have been hired as a junior **network technician** to design a small office network for a company called DataBridge Innovations.

The office has three departments:

- **Admin Department** – handles internal operations
- **Sales Department** – handles customer communication
- **HR Department** – handles recruitment tasks

All departments must be connected so that devices can communicate across the network.

The company wants the network designed and tested using **Cisco Packet Tracer**, ensuring that:

- All devices are correctly connected
- IP addresses are properly assigned
- Basic connectivity is confirmed using simple network tests



The Business Problem

DataBridge Innovations is expanding its operations and requires a reliable internal network to support communication across its three key departments: Admin, Sales, and HR. Currently, there is no structured network infrastructure, which limits collaboration, slows down operations, and increases the risk of miscommunication and inefficiency.

The Problem

The company lacks a properly designed Local Area Network (LAN) that enables seamless communication between departments. Without this:

- Devices cannot communicate effectively across departments
- There is no proper IP addressing scheme
- Network performance and reliability cannot be guaranteed
- Troubleshooting and scalability become difficult

Your Role

As a **Junior Network Technician**, your responsibility is to:

- Design a functional network topology using Cisco Packet Tracer
- Connect all departmental devices appropriately (PCs, switches, routers if needed)
- Assign correct IP addresses to all devices
- Ensure inter-department communication is possible
- Test and verify connectivity using basic tools like ping



Rationale for the Project

This Project should....

Rationale 1



Improve Communication Efficiency

A structured network ensures that Admin, Sales, and HR can share information quickly and reliably, improving overall workflow.

Rationale 2



Enhance Network Organization

Segmenting departments logically (even in a small network) allows better management, easier troubleshooting, and future scalability.

Rationale 3



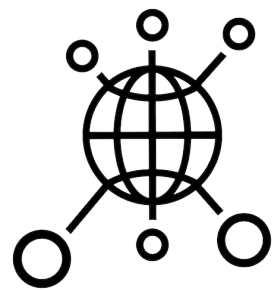
Support Business Growth

A properly designed network provides a foundation for future expansion, such as adding more users, devices, or services.



Project Initiative

By completing this project, you will be able to:



Network Design & Topology Creation

- Create a simple LAN topology using switches to connect all departments
- Ensure proper physical and logical layout



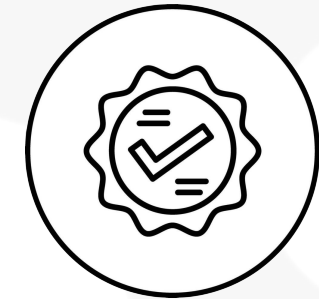
Device Configuration

- Assign IP addresses to all devices using a consistent addressing scheme
- Configure basic settings on devices



Connectivity Implementation

- Ensure all devices are properly connected using appropriate cables
- Enable communication between departments



Testing & Validation

- Use ping tests to confirm connectivity
- Troubleshoot any communication failures
- Validate that all departments can communicate successfully

Network Overview

Department	Network/Subnet	Devices
Admin	192.168.10.0/24	2 PCs & 1 server
Sales	192.168.20.0/24	2 PCs
HR	192.168.30.0/24	2 PCs

Router(2911): Connects all three networks

Switches(2960): One per department

Project Workflow

Step 1: Network Setup

- Open Cisco Packet Tracer.
- Add 1 Router(2911), 3 Switches, and 6 PCs (2 per department).
- Connect the devices using straight-through cables.

Step 2: IP Address Configuration

- Assign IP addresses to each PC and corresponding router interfaces.

Example:

- Admin: 192.168.10.1 (router), 192.168.10.10–10.11 (PCs)
- Sales: 192.168.20.1 (router), 192.168.20.10–20.11 (PCs)
- HR: 192.168.30.1 (router), 192.168.30.10–30.11 (PCs)

Step 3: Test Connectivity

Use the ping command between departments.

- At this stage, all PCs should communicate successfully.

→ Expected Outcome



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At the end of this practical, participants will understand:

- How a basic routed network works
- How IP addressing enables communication
- How to test and troubleshoot simple network Connectivity



PROJECT (ASSIGNMENT)

Note: This assignment introduces ACLs in a **very basic way**. No complex rules.

Task

1. Using the same network you already built:
 - Apply one simple ACL rule on the router
2. Configure the ACL so that:
 - HR Department (192.168.30.0/24) cannot ping the Admin Department (192.168.10.0/24)
 - All other traffic should be allowed



What to Submit

Students should submit:

- Screenshot showing:
 - Successful ping from Sales to Admin
 - Failed ping from HR to Admin
- Screenshot of the ACL command used

Let's Dive in

Please note

When you are done with this case study:
Don't forget to **add it to your portfolio** or
share on LinkedIn

Remember to tag **@10Alytics** and use
these #Networksecurity
#Networking
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